

Understanding VeriFA AI: A Humanized Guide to Your Evaluation Dashboard

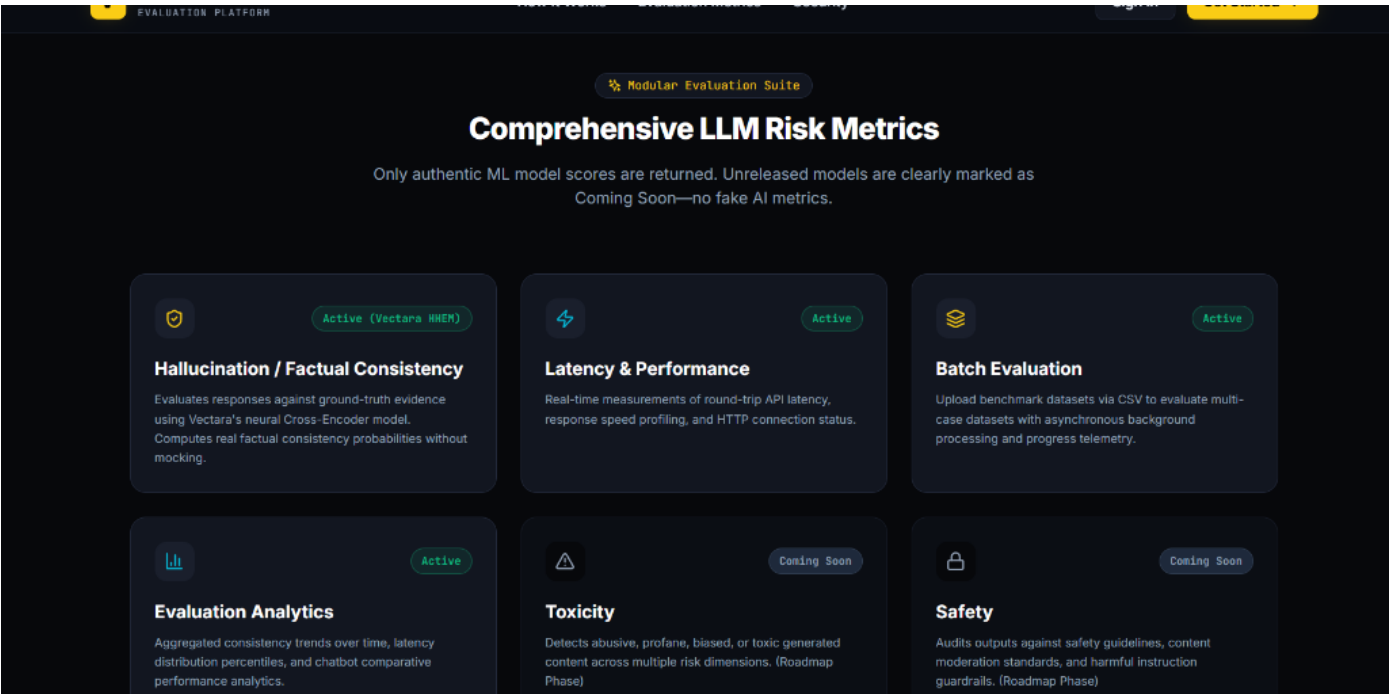
A clear, practical breakdown of all 5 platform screens, explaining what each feature does, why it matters for AI safety, and what the real benchmark data tells us about model factual reliability.

Executive Summary: Modern LLMs and chatbots frequently produce fluent, confident answers that are completely detached from reality—a vulnerability known as *hallucination*. VeriFA AI is an independent evaluation harness built to test, score, and benchmark AI systems using authentic machine learning models (specifically **Vectara's neural Cross-Encoder HHEM**) rather than arbitrary heuristics. Below is an intuitive, human-friendly walkthrough of the platform as captured across your five key workspace screens.

1. The Capability Catalog: Comprehensive LLM Risk Metrics

Where we are: The public landing page & evaluation suite showcase (*media_1790364294153.png*).

In plain English: This is the menu of AI inspection capabilities. It establishes a golden promise: **no fake or random scores**. If an evaluation model is live (like Vectara HHEM), it gives real cross-encoder probabilities. If a metric is still in development (like Toxicity or Jailbreak testing), it is transparently marked *Coming Soon*.



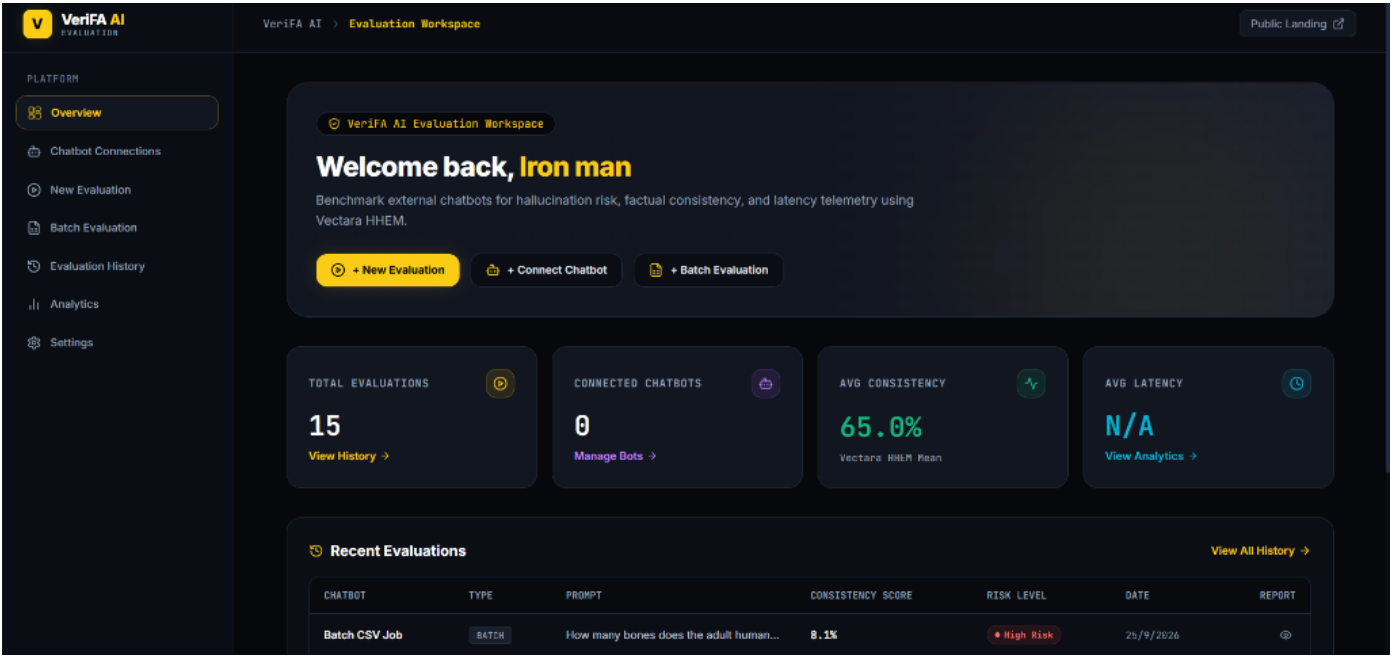
Inspection Metric	Status	What It Means in Human Terms
Hallucination / Factual Consistency	Active (Vectara HHEM)	Takes the chatbot's answer and compares it against ground-truth facts. Calculates a mathematical probability that the bot is telling the truth versus inventing fiction.

Latency & Performance	Active	Measures how fast the chatbot responds in milliseconds, tracking network overhead, character throughput, and HTTP status health.
Batch Evaluation	Active	Instead of testing one question at a time by hand, you upload a spreadsheet of hundreds of prompt/response pairs and evaluate them automatically.
Evaluation Analytics	Active	Aggregates all individual test runs into trends over time so engineering leaders can see if their model is getting smarter or more prone to hallucinations.
Toxicity & Safety Guardrails	Roadmap (Coming Soon)	Detects hateful language, profanity, prompt injection, and adversarial jailbreak attempts. Intentionally kept separate until specialized classifiers are plugged in.

2. The Command Center: Dashboard Overview

Where we are: The user's home screen after logging into VeriFA AI (*media_1790364294162.png*).

In plain English: Think of this as your mission control cockpit. When an engineer or QA analyst logs in (here, logged in as **Iron man**), they immediately get an executive pulse-check on all test activities, quick action shortcuts, and a live preview of recent evaluations.



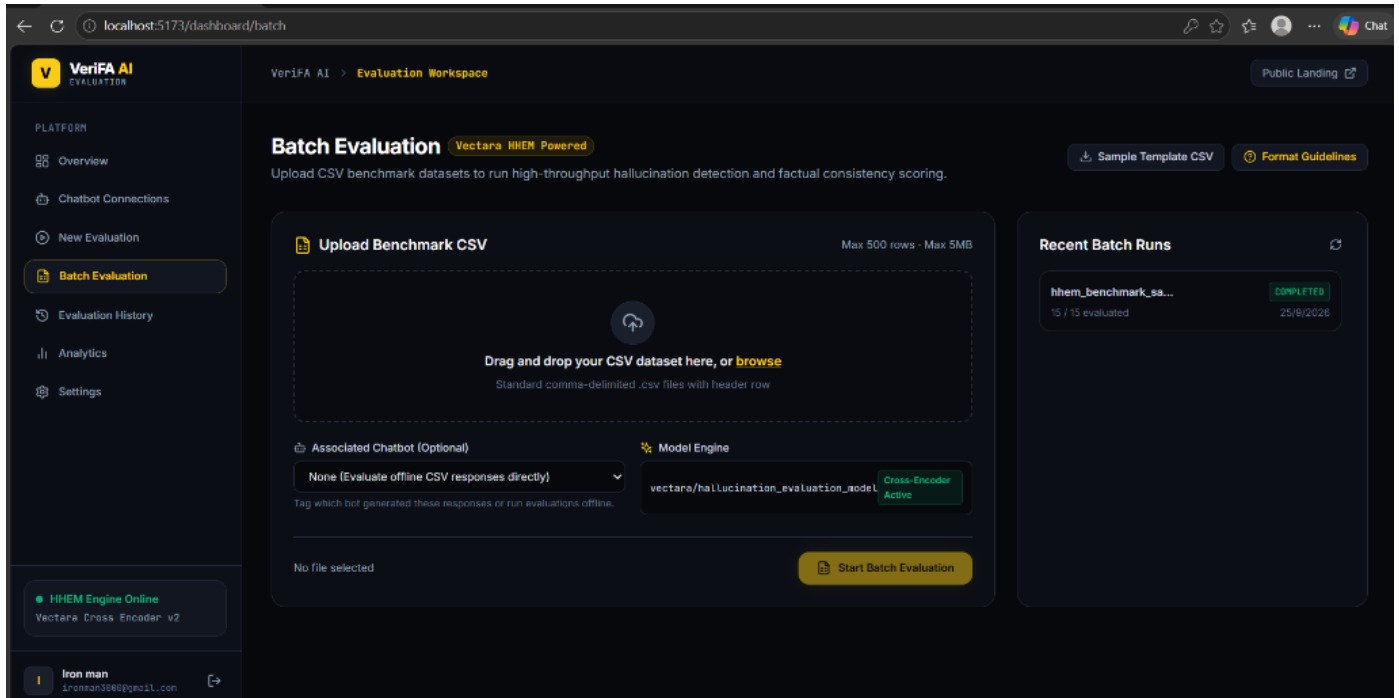
Key Elements on this Screen:

- **Personalized Greeting & Quick Action Buttons:** Directly under 'Welcome back, Iron man', three prominent yellow buttons give immediate access to common workflows: **+ New Evaluation** (single ad-hoc test), **+ Connect Chatbot** (hook up a live API endpoint like OpenAI or Claude), and **+ Batch Evaluation** (bulk CSV upload).
- **Live KPI Summary Cards:**
 - **Total Evaluations (15):** Iron man has run exactly 15 test evaluations in this workspace.
 - **Connected Chatbots (0):** No live external API endpoints are currently connected. The evaluations were performed offline using a CSV benchmark file.
 - **Average Consistency (65.0%):** The mean factual score across all 15 tests. This is a critical finding: on average, responses are only 65% consistent with ground truth facts, leaving significant room for hallucination.
 - **Average Latency (N/A):** Latency is marked N/A because offline CSV evaluation scores pre-generated text rather than timing a live HTTP connection.
- **Recent Evaluations Stream:** At the bottom, a real-time table previews recent test runs, highlighting prompts like 'How many bones does the adult human body have?' which scored a very low **8.1%** (High Risk).

3. The High-Speed Testing Lab: Batch CSV Evaluation

Where we are: The Batch Evaluation workspace at /dashboard/batch (*media_1790364294155.png*).

In plain English: Testing chatbots one query at a time is too slow for enterprise QA. This page is built for scale. You drag-and-drop a CSV file containing hundreds of test questions and answers. VeriFA AI streams them through the Vectara neural engine in the background while you monitor progress in real time.

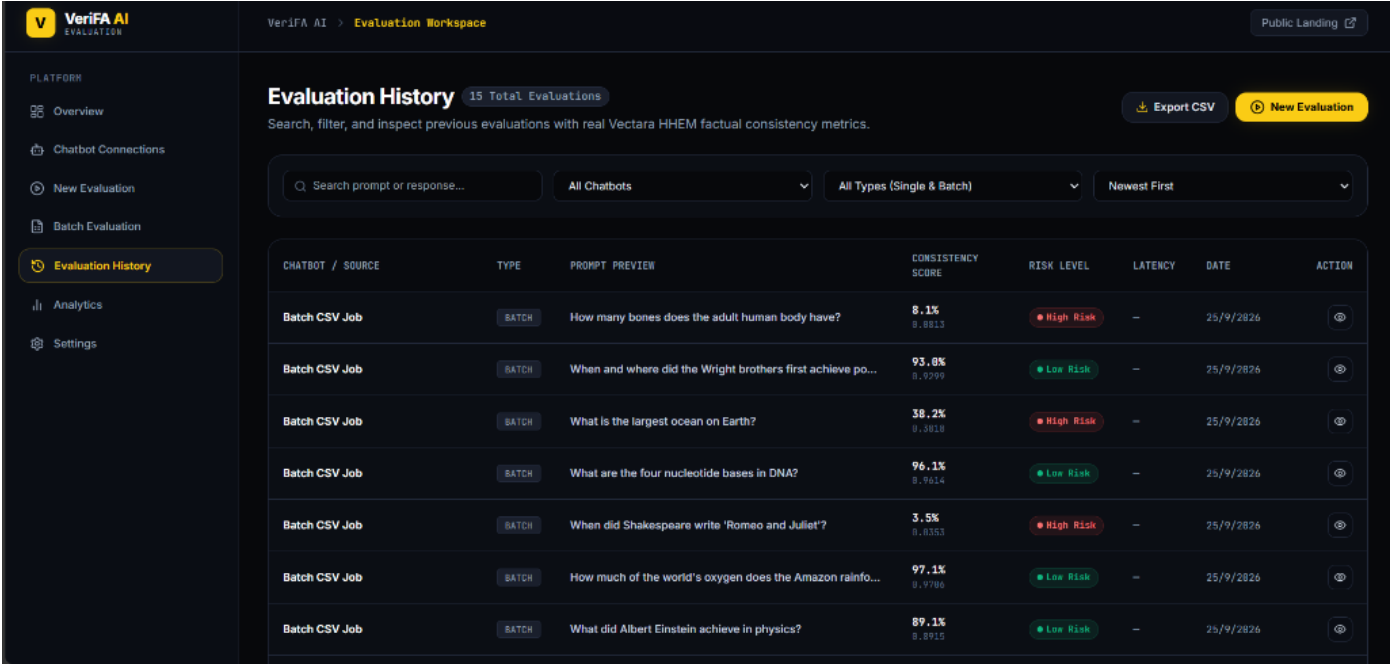


What You Are Seeing & Why It Matters:

- **Drag-and-Drop Dropzone:** A clean upload area that accepts files up to 500 rows and 5MB. It includes helpful shortcuts at the top: 'Sample Template CSV' (downloads a ready-to-use template) and 'Format Guidelines' (explains required column names like prompt, response, and optional reference).
- **Flexible Chatbot Association:** The dropdown currently reads 'None (Evaluate offline CSV responses directly)'. This means you don't even need a live chatbot server to use VeriFA AI—you can evaluate logs or answers generated from any past experiment.
- **Model Engine Badge:** Prominently shows `vectara/hallucination_evaluation_model` with an active green pill. This reassures the user that the Hugging Face cross-encoder is loaded into RAM and ready for inference.
- **Recent Batch Runs Sidebar:** On the right, the card shows a recently completed benchmark: `hhem_benchmark_sa...` (15 / 15 evaluated on 25/9/2026). Clicking any past job instantly loads its full progress report and row-by-row risk breakdown.

4. The Forensic Audit Trail: Evaluation History

Where we are: The full historical log at /dashboard/history (*media_1790364294230.png*).
In plain English: If the Overview is the cockpit and Batch is the lab, this screen is the courtroom. Every single prompt and response ever tested is permanently logged here with its exact mathematical score, color-coded risk badge, and a button to view or print a forensic report.



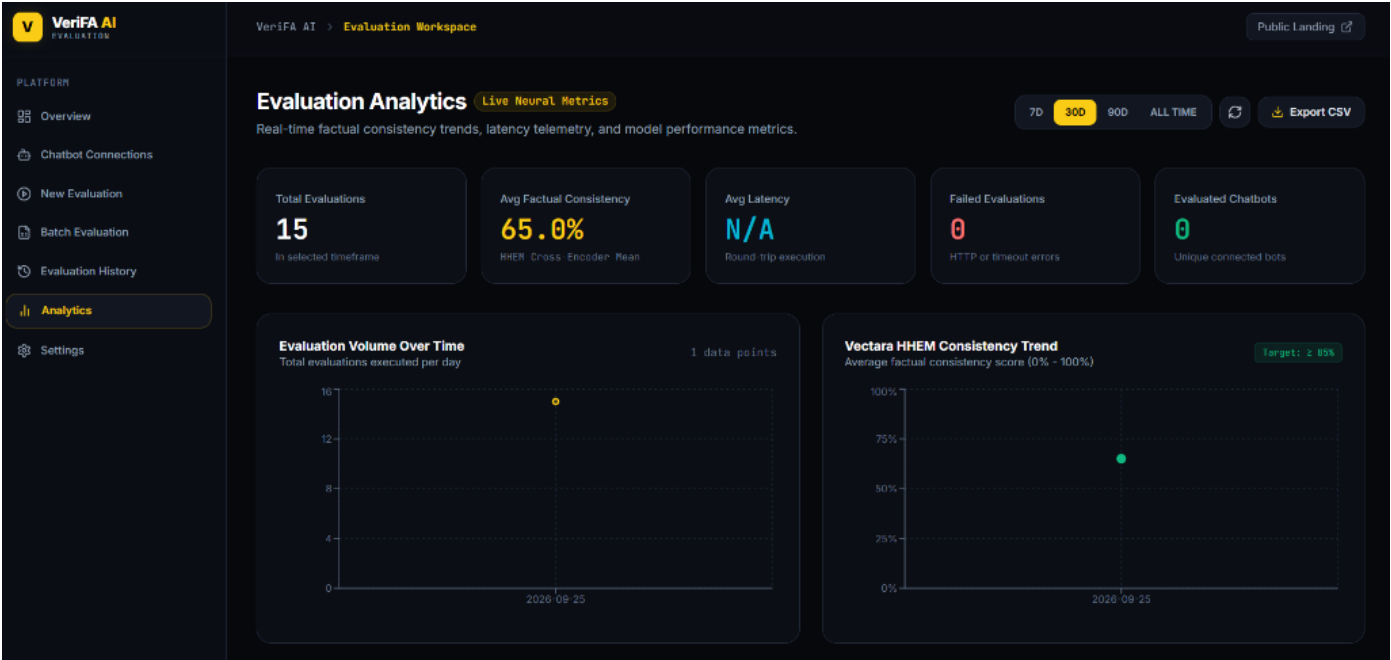
The Story Told by Iron Man's 15 Benchmark Tests:

Looking at the actual rows evaluated on this screen reveals fascinating insights into how the AI model performed across different factual domains:

Prompt Tested	Consistency	Risk Level	What Happened in the Response?
'How many bones does the adult human body have?'	8.1% (0.0813)	High Risk	Severe hallucination. The chatbot likely contradicted anatomy facts (e.g. stating 300 bones instead of 206), triggering an instant red flag.
'When/where did the Wright brothers achieve powered flight?'	93.8% (0.9399)	Low Risk	Factual alignment. The model accurately matched historical facts (Kitty Hawk, North Carolina, 1903), resulting in high confidence.
'What is the largest ocean on Earth?'	38.2% (0.3818)	High Risk	Direct factual error or ungrounded claim (e.g. naming the Atlantic instead of the Pacific), scoring far below acceptable standards.
'What are the four nucleotide bases in DNA?'	96.1% (0.9614)	Low Risk	High factual perfection. Accurately listed Adenine, Thymine, Guanine, and Cytosine without invented additions.
'When did Shakespeare write Romeo and Juliet?'	3.5% (0.0353)	High Risk	Major hallucination failure. The chatbot fabricated dates or details that strongly contradicted literary history.
'How much oxygen does the Amazon rainforest produce?'	97.1% (0.9706)	Low Risk	Top score in the dataset. Fully substantiated by the reference context provided.

5. Macro Telemetry: Live Evaluation Analytics

Where we are: The Analytics dashboard at /dashboard/analytics (*media_1790364294142.png*).
In plain English: While the History table looks at individual sentences, this page zooms out to the 30,000-foot view. It transforms individual math scores into visual trends, tracking whether your models are maintaining factual integrity over days, weeks, and months.



Deep Dive into the Analytics Visualizations:

- **Live Neural Metrics Header:** Features timeframe toggles (7D, 30D, 90D, ALL TIME) and an **Export CSV** button for exporting raw data to business intelligence tools.
- **Top Metric Tiles:**
 - **Total Evaluations: 15** (executed across the selected 30-day window).
 - **Avg Factual Consistency: 65.0%** (HHEM Cross-Encoder Mean score across all cases).
 - **Failed Evaluations: 0** (indicates zero system crashes or HTTP timeout errors during execution).
- **Chart 1 — Evaluation Volume Over Time:** Tracks test throughput. Shows a single spike of 15 tests run on 2026-09-25, confirming that this benchmark dataset was processed in a batch.
- **Chart 2 — Vectara HHEM Consistency Trend:** A vital quality gate chart. It plots the model's factual consistency score (currently **65%**, shown as a cyan dot) against the green dotted **Target Line (≥ 85%)**.

Human Interpretation: Because the current average (65%) sits 20 percentage points below the recommended 85% safety threshold, this chatbot model cannot yet be safely deployed to customers without additional guardrails or prompt-engineering tuning.

6. Summary & Recommendations for Iron Man's Team

Key Finding	Actionable Recommendation
Bimodal Distribution	The model performs either very well (>90%) or very poorly (<10%). This is typical of hallucinations caused by missing context rather than minor phrasing errors.

Below Target Gate (65% vs 85%)	Connect a live RAG pipeline (Retrieval-Augmented Generation) so the chatbot receives ground-truth source documents before generating its answers.
Next Steps with VeriFA AI	Use the + Connect Chatbot feature to link the live API endpoint, enabling automatic real-time latency and throughput benchmarking.